



Waymo v. Uber (2017) Insider Threats, Trade Secret Theft & Corporate Governance

**MIT 5600G: Security Policies and
Risk Management**

Case Study : 30

Group Members — Shah Md Julhash Tamim
(101002572)

Faishal Youshuf (101044804)

**SUBMISSION DATE : 22 JUNE,
2026**

Presentation Overview



Background: Organizations & Incident

Attack Timeline: How the Incident Unfolded

Technical Aspects: The Stolen Trade Secrets

Organizational & Governance Failures

Incident Response Evaluation

Legal, Ethical & Business Implications

Recommendations & Lessons Learned

Connection to Class Concepts

Discussion Questions



Background of the Organizations



Waymo: A subsidiary of Alphabet Inc. (Google's parent company) focused on autonomous vehicle (AV) development. Waymo invested over \$1.1 billion and years of intensive R&D into mastering autonomous technology.



Uber Technologies: A global ride-hailing giant that viewed autonomous driving as vital to its long-term corporate survival, attempting to build a competing program rapidly.

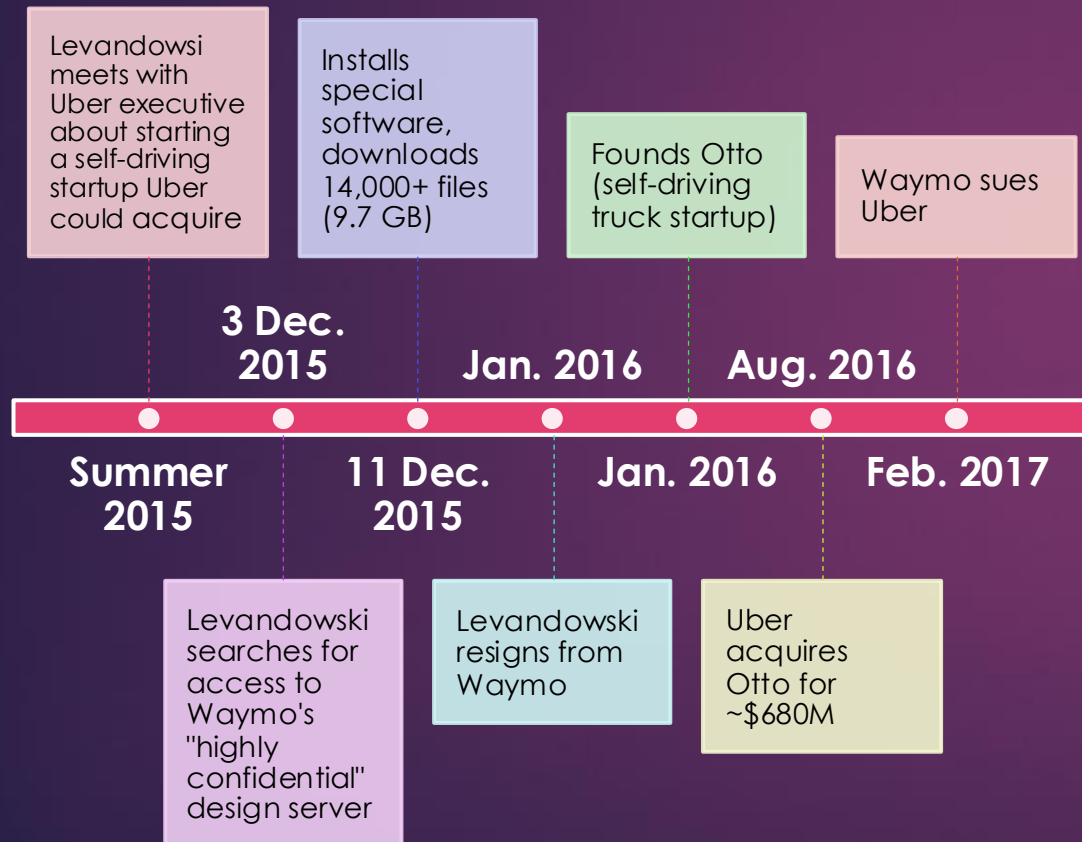
1. Case Background

Background – The Key Individuals

- ▶ **The Central Figure: Anthony Levandowski**
- ▶ **Anthony Levandowski**
- ▶ Senior engineer / manager on Waymo's self-driving car project
- ▶ Deeply involved in Waymo's LiDAR technology development
- ▶ Left Waymo in **January 2016** to found Otto
- ▶ **The Allegation:**
- ▶ Downloaded **over 14,000** confidential files before resigning
- ▶ Transferred **9.7 GB** of proprietary Waymo data to an external hard drive
- ▶ Used stolen IP to launch Otto, which Uber then acquired



Incident Timeline

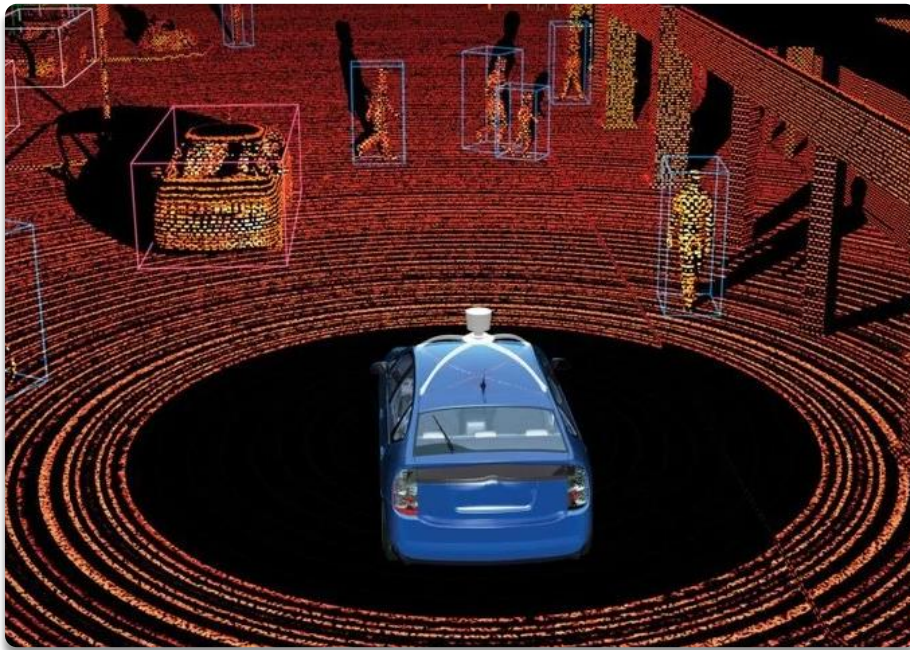


Legal Proceedings

Feb. 2017	Waymo files lawsuit: trade secret misappropriation, patent infringement, unfair competition
April 2017	Levandowski invokes Fifth Amendment , refuses to testify
May 2017	Judge Alsup grants limited preliminary injunction against Uber
May 2017	Judge refers case to U.S. Department of Justice for criminal investigation
Sept 2017	Judge denies Uber's motion to move case to arbitration
Feb. 2018	Settlement reached mid-trial — Uber pays \$245M in equity



Technical Aspects: LiDAR (stolen technology)



Source: google

- ▶ **What is LiDAR?**
- ▶ **Light Detection and Ranging** — the "electronic eyes" of autonomous vehicles
- ▶ Enables self-driving cars to detect traffic, pedestrians, and obstacles
- ▶ Each manufacturer develops proprietary, unique LiDAR systems
- ▶ **What Levandowski Took:**
- ▶ 14,000+ confidential files
- ▶ Technical **blueprints, design files, schematics**
- ▶ **Circuit board designs** (Trade Secret 14: positioning guide pins)
- ▶ System architectures and specifications

Technical Aspects: The Evidence

- ▶ **How Was the Theft Discovered?**
- ▶ **The "Smoking Gun" Email**
- ▶ Waymo employee was **inadvertently copied** on an email from a LiDAR supplier
- ▶ Email contained **circuit board drawings** eerily similar to Waymo's proprietary designs
- ▶ **Forensic Evidence**
- ▶ Forensic analysis confirmed **14,000+ files** were transferred to external storage
- ▶ Levandowski accessed Waymo's SVN schematic repository on **Dec 11, 2015**
- ▶ Downloaded data included **9.7 GB** of confidential information

Organizational Failures and Incident Response

► Organizational Failures

- No restrictions on mass downloads – Levandowski had legitimate, unrestricted access
- No real-time DLP alerts for abnormal file transfers
- Sensitive files were not persistently encrypted
- Documents lacked automatic classification as trade secrets
- Offboarding controls did not prevent data exfiltration before resignation
- No behavioral monitoring correlated download activity with risk context

► Incident Response

- Waymo conducted an internal investigation, filed suit promptly (Feb 2017), obtained a preliminary injunction, and referred the case to the DOJ
- Uber initially denied using Waymo technology, failed to disclose vital evidence, and demoted Levandowski only after legal pressure
- Settlement reached mid-trial: Uber paid **\$245M** in equity
- Levandowski was charged with **33 counts** and sentenced to **18 months** in federal prison (later pardoned)

Legal, Ethical and Business Impact

► Legal Consequences

- Civil suit under the **Defend Trade Secrets Act** and **California Uniform Trade Secrets Act** – settled for **\$245M**
- Criminal charges: **33 counts** of trade secret theft – Levandowski received **18 months** in prison
- Judge Alsup referred the case to the **DOJ**, which also triggered **shareholder lawsuits** against Uber's board

► Ethical Issues

- Did Uber knowingly hire an engineer who had stolen trade secrets?
- Levandowski's **Fifth Amendment** invocation raised questions of guilt
- Corporate culture that prioritized speed over ethical conduct

► Business Impact

- Reputational damage to Uber's brand and leadership
- Uber's self-driving program was **delayed by years**
- Financial toll: **\$245M settlement** + legal fees + lost competitive advantage
- Waymo's ~**\$1.1B** R&D investment was partially devalued

Journal of Business Cases and Applications

Volume 27

The Case of Waymo V. Uber: The Detrimental Harm to the Evolution of Trade Secret Laws

Racquel Brown Gaston, Esq.
University of the District of Columbia

J. Ndumbe Anyu, Ph.D.
University of the District of Columbia

Abstract

In a high-speed chase toward being the first innovator in an ever-changing infusion of business and technology, crashes of technological giants' intellectual property are almost certain to take place yet again, in comparison to their unprecedented progression, found themselves at the mercy of the one entity that many critics have said seemed to have stood still in time - the law. Oftentimes criticized for its inability to keep up with society's swift technological advancements, it is this *slow* human component of the law in contrast to the unpredictable evolution of a robotics society, that has *ensured* safeguards, thus necessitating rulings for the necessary protection of these same innovators. In one's over sweeping critique of the law's pace in contrast to technology, it is mindful to note that such critique cannot apply to all four sources of the law. Legislative bodies and administrative agencies have a responsibility to be vigilant and intentional in predicting technological shifts and enact safeguards or face the consequences synonymous with that of the wild-west. On the other hand, the Constitution, is the bedrock for the United States' governing structure and the fundamental laws for its citizens. Those have been cemented to withstand the test of time and thus must move as slowly as a continental shift to maintain consistency and reliability is an ever-changing society.

As for case law, it has always evolved from crisis. It is from the agony of disputes that this law has been shaped. It cannot be ahead of society as it is shaped in the here and now of its citizens. This is evident in the fact that establishing standing-injury sustained, as oppose to injury one thinks will be sustained is a requirement for judicial intervention regarding resolving disputes. The critique therefore that case law is ill equipped to meet the challenges of innovation is only correct to the extent that such disputes, evidenced by Waymo v. Uber are either not resolved by the courts, based on the parties' decision-hence the non-creation of case law in that particular field, or parties do not believe that disputes have risen to the level where judicial intervention is required. This paper will use the settled 2018 case of Waymo v. Uber to challenge the over sweeping theory that the law is not keeping at pace with the evolutionary speed of technology, thus implying that newly sprouted issues will likely leave courts in the position comparable to that of deer in headlights.

Keywords: Waymo, Uber, Google, Alphabet, Intellectual Property, Self-Driving Cars, Lidar, Trade Secrets, Anthony Levandowski

Copyright statement: Authors retain the copyright to the manuscripts published in AABRI journals. Please see the AABRI Copyright Policy at <http://www.aabri.com/copyright.html>

The case of Waymo, Page 1

Lessons Learned: Access Control

- ▶ **Lesson 1: Strengthen Access Controls**
- ▶ **Principle of Least Privilege**
 - ▶ Restrict access to IP strictly to those who need it
 - ▶ Regularly audit permissions and rotate credentials
- ▶ **Monitor Data Activity**
 - ▶ Track abnormal file transfers, especially before resignations
 - ▶ Correlate download behavior with risk context (e.g., resignation notices)
- ▶ **Segmentation & Encryption**
 - ▶ Persistently encrypt sensitive files
 - ▶ Implement network segmentation to limit data exfiltration pathways



DLP, Monitoring, Offboarding & Hiring

► DLP & Monitoring

- Auto-classify and encrypt sensitive files
- Behavioral analytics for abnormal mass downloads
- Real-time alerts for data transfers before resignations
- Prioritize monitoring of high-risk IP-access employees

► Offboarding & Hiring

- Revoke access immediately upon termination
- Monitor activity during notice periods
- Conduct exit interviews; collect all devices
- Vet competitor hires for trade secret exposure
- Enforce strong NDAs, non-competes, and IP clauses

► Short Example:

Waymo lacked a DLP rule to block mass USB transfers. A simple policy restricting external drives would have stopped the **9.7 GB** exfiltration instantly.



Course Connections & Recommendations

Insider Threats –
Malicious insider with
fully authorized
access

Access Control –
Failure of least
privilege and network
segmentation

Data Classification –
No automatic
tagging of files as
trade secrets

Incident Response –
Detection occurred
only via a vendor
email (major gap)

Governance – Uber's
board approved
acquisition without
proper IP due
diligence

Risk Management –
Overlooked insider
risk and third-party
hiring risks

**Key
Recommendations**

Zero Trust – Verify
every access request,
regardless of user role

AI-Driven DLP –
Deploy behavioral
analytics to detect
anomalies in real time

**Continuous
Monitoring** – Prioritize
departing employees
and high-risk users

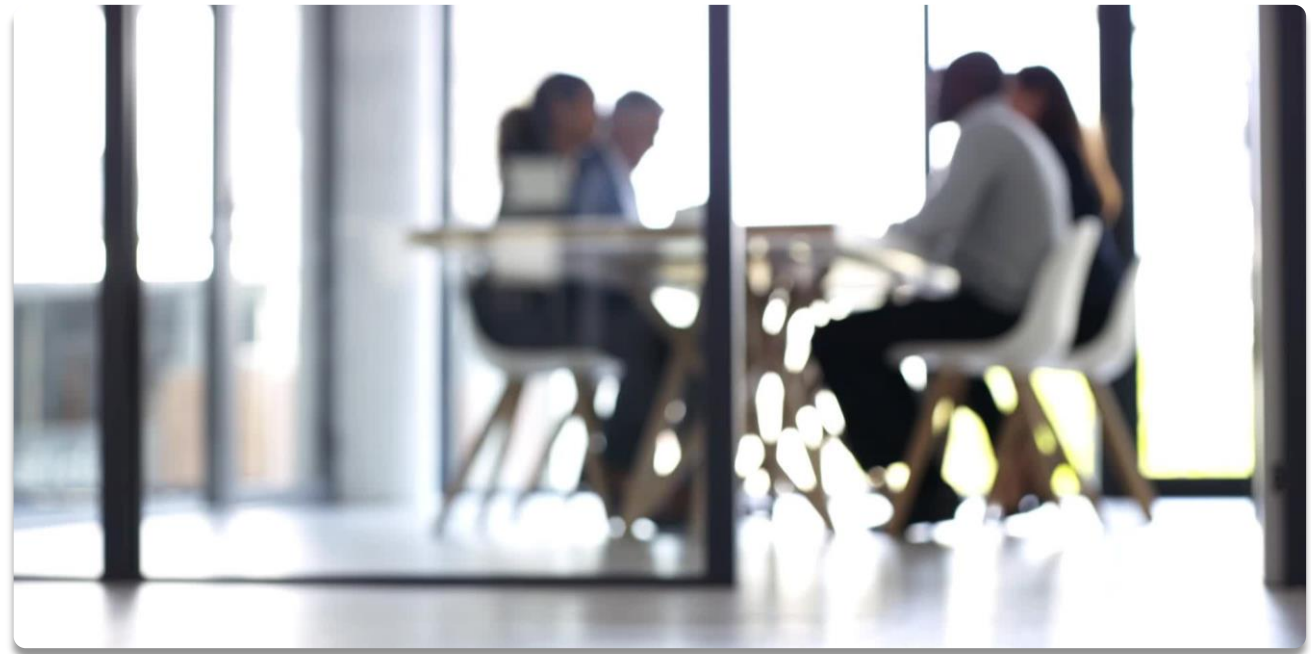
Secure Offboarding –
Automate access
revocation and
device collection

Legal Safeguards –
Strengthen NDAs,
non-competes, and
IP ownership clauses

Culture – Foster
security awareness
with regular,
role-specific training

Discussion Questions /1

1. If you were leaving a company, would you expect your access to sensitive files to be restricted or monitored?
2. Should companies have the right to monitor everything an employee does on their work devices?
3. Is it fair that Uber had to pay \$245 million even though Levandowski was the one who stole the files?



Discussion Question /2

4. What would you do if you discovered a new hire brought confidential files from their previous employer?

5. How much monitoring is too much monitoring in the workplace? When does it cross into invading privacy?

Thank You
